Grassmannians and the Segre Embedding

Adeetya Choubey - 22014

BS-MS '22

Abstract

This article first establishes the Grassmanian, $\mathrm{Gr}(d, V)$, V being an n dimensional vector space over the field k, as a projective variety, via the Plücker embedding, and shows that it is a projective algebraic set by looking for defining polynomials. We work through the specific example of $\mathrm{Gr}(2, k^4)$. We then turn our attention to showing that product of two projective spaces is a projective variety, with the help of segre map.

Contents

---

1 The Grassmannian as a Projective Variety

1.1 The setup

Let k be an algebraically closed field and V a k-vector space of dimension n. The Grassmanian, $\mathrm{Gr}(d, V)$ is the collection of all the d - dimensional subspaces of the n - dimensional vector space over our field of choice. Fix a basis $\{e_1, \dots, e_n\}$ of V. The induced basis for $\bigwedge^d V$ is $\{e_{i_1} \wedge \cdots \wedge e_{i_d} \mid 1 \le i_1 < \cdots < i_d \le n\}$, giving $\dim \bigwedge^d V = \binom{n}{d}$. We set $N = \binom{n}{d} - 1$ and identify $\mathbb{P}(\bigwedge^d V) \cong \mathbb{P}^N$. Denote by p_I the homogeneous coordinate corresponding to $e_{i_1} \wedge \cdots \wedge e_{i_d}$ for $I = (i_1, \dots, i_d)$.

1.2 The Plücker Embedding

Definition 1.1. The *Plücker map* is

$$p\colon \mathrm{Gr}(d,V) \longrightarrow \mathbb{P}(\bigwedge^d V), \qquad W \longmapsto [w_1 \wedge \cdots \wedge w_d],$$

where $\{w_1,\dots,w_d\}$ is any basis of $W \in \mathrm{Gr}(d,V)$.

If $\{v_1,\dots,v_d\}$ is another basis related to $\{w_j\}$ by $v_j = \sum_i a_{ij} w_i$, then $v_1 \wedge \cdots \wedge v_d = \det(a_{ij})\, w_1 \wedge \cdots \wedge w_d$ (Lemma 1.2 of [6]), so the projective class is independent of the chosen basis and p is well defined.

Proposition 1.2. *The Plücker map $p\colon \mathrm{Gr}(d,V) \to \mathbb{P}(\bigwedge^d V)$ is injective.*

Proof. Define $\varphi([\omega]) = \{v \in V \mid v \wedge \omega = 0 \in \bigwedge^{d+1} V\}$, which is a subspace of V. For W with basis $\{w_1,\dots,w_d\}$, every $w \in W$ satisfies $w \wedge w_1 \wedge \cdots \wedge w_d = 0$, so $W \subseteq \varphi(p(W))$. Conversely, extend $\{w_i\}$ to a basis $\{w_1,\dots,w_n\}$ of V and write $v = \sum a_i w_i$. Then

$$0 = v \wedge w_1 \wedge \cdots \wedge w_d = \sum_{j>d} a_j\, w_j \wedge w_1 \wedge \cdots \wedge w_d,$$

and since the terms $w_j \wedge w_1 \wedge \cdots \wedge w_d$ $(j>d)$ are linearly independent in $\bigwedge^{d+1} V$, all $a_j = 0$ for $j > d$. Hence $v \in W$, giving $\varphi(p(W)) \subseteq W$, so $\varphi \circ p = \mathrm{id}$ and p is injective. □

1.3 Totally Decomposable Vectors

Definition 1.3. $\omega \in \bigwedge^d V$ is *totally decomposable* if $\omega = v_1 \wedge \cdots \wedge v_d$ for some linearly independent $v_1,\dots,v_d \in V$.

For $\omega \in \bigwedge^d V$ write $D_\omega = \{v \in V \mid v \wedge \omega = 0\}$.

Lemma 1.4. *$v \in V$ divides $\omega \in \bigwedge^d V$ (i.e. $v \wedge \omega = 0$) if and only if $v = 0$ or $\omega = \varphi \wedge v$ for some $\varphi \in \bigwedge^{d-1} V$.*

Proof. The direction $(\Leftarrow)$ is immediate. For $(\Rightarrow)$, choose a basis $\{v, u_2, \dots, u_n\}$ of V and write $\omega = \sum_I a_I\, e_I$ in the induced basis for $\bigwedge^d V$. From $v \wedge \omega = 0$, all terms a_I with $1 \notin I$ must vanish (as those wedge products are independent), so $\omega = v \wedge \left(\sum_{I \ni 1} a_I\, e_{I \setminus \{1\}}\right)$. □

Lemma 1.5. *$\omega \in \bigwedge^d V$ is totally decomposable if and only if* $\dim D_\omega = d$.

Proof. $(\Rightarrow)$ Let $\omega = v_1 \wedge \cdots \wedge v_d$. Clearly $v_1,\dots,v_d \in D_\omega$ are linearly independent. For any $v = \sum a_i w_i$ (extending $\{v_j\}$ to a basis), $v \wedge \omega = 0$ forces $a_j = 0$ for $j > d$ by the same independence argument as in Proposition 1.2. Thus $D_\omega = \mathrm{span}\{v_1,\dots,v_d\}$.

$(\Leftarrow)$ Let $\{v_1,\dots,v_d\}$ be a basis of D_ω; extend to $\{v_1,\dots,v_n\}$. Since $v_j \wedge \omega = 0$ for each $j \le d$, Lemma 1.4 applied iteratively forces all basis coefficients of ω whose multi-index does not equal $\{1,\dots,d\}$ to vanish. Hence $\omega = \lambda\, v_1 \wedge \cdots \wedge v_d$ for some scalar $\lambda \neq 0$. □

Lemma 1.6. $[\omega] \in \mathbb{P}(\bigwedge^d V)$ *lies in* $p(\mathrm{Gr}(d, V))$ *if and only if* ω *is totally decomposable.*

Proof. If $\omega = v_1 \wedge \cdots \wedge v_d$, then $W = \mathrm{span}\{v_i\} \in \mathrm{Gr}(d, V)$ and $p(W) = [\omega]$. Conversely, $p(W) = [\omega]$ implies $\omega = \lambda\, w_1 \wedge \cdots \wedge w_d$ for a basis $\{w_i\}$ of W. □

1.4 The Rank Condition and the main theorem

Definition 1.7. For $\omega \in \bigwedge^d V$, the *contraction map* is

$$\varphi_\omega : V \longrightarrow \bigwedge^{d+1} V, \qquad v \longmapsto \omega \wedge v.$$

Lemma 1.8. ω *is totally decomposable if and only if* $\mathrm{rank}(\varphi_\omega) \leq n - d$.

Proof. By definition, $\ker \varphi_\omega = D_\omega$. By Lemma 1.5, ω is totally decomposable iff $\dim D_\omega = d$ iff $\mathrm{rank}\, \varphi_\omega = n - d$. The rank never drops below $n-d$: if $\dim D_\omega = l > d$, then applying the ($\Leftarrow$) direction of Lemma 1.5 to the l-dimensional kernel would express $\omega \in \bigwedge^d V$ as a scalar multiple of a d-fold wedge from a basis of D_ω, yet $l > d$ means only a d-element sub-basis is needed—so ω is already totally decomposable, contradicting $l > d$. Thus $\mathrm{rank}\, \varphi_\omega \geq n - d$ always, and the condition $\leq n - d$ is equivalent to $= n - d$. □

Theorem 1.9. *Under the Plücker embedding,* $p(\mathrm{Gr}(d, V)) \subset \mathbb{P}(\bigwedge^d V)$ *is the projective algebraic set defined by the vanishing of all* $(n - d + 1) \times (n - d + 1)$ *minors of the matrix of* φ_ω*; in particular, it is a projective variety.*

Proof. Consider the assignment map $\Phi\colon \bigwedge^d V \to \mathrm{Hom}(V, \bigwedge^{d+1} V)$ defined by $\omega \mapsto \varphi_\omega$. For any scalars $c_1, c_2 \in k$ and tensors $\omega_1, \omega_2 \in \bigwedge^d V$, evaluating the action on an arbitrary $v \in V$ gives:

$$\begin{aligned} \Phi(c_1\omega_1 + c_2\omega_2)(v) &= (c_1\omega_1 + c_2\omega_2) \wedge v \\ &= c_1(\omega_1 \wedge v) + c_2(\omega_2 \wedge v) \\ &= c_1\varphi_{\omega_1}(v) + c_2\varphi_{\omega_2}(v). \end{aligned}$$

Since this holds for all $v \in V$, we have $\Phi(c_1\omega_1 + c_2\omega_2) = c_1\Phi(\omega_1) + c_2\Phi(\omega_2)$, proving Φ is a linear map.

Let us explicitly construct the matrix M_ω representing φ_ω. The columns of M_ω are indexed by the basis vectors e_k of V, and the rows are indexed by the basis vectors e_J of $\bigwedge^{d+1} V$. Writing $\omega = \sum_{|I|=d} p_I e_I$, we find the k-th column of M_ω by applying φ_ω to e_k:

$$\varphi_\omega(e_k) = \omega \wedge e_k = \sum_{|I|=d} p_I (e_I \wedge e_k).$$

By the properties of the wedge product, if $k \in I$, $e_I \wedge e_k = 0$. If $k \notin I$, $e_I \wedge e_k = \pm e_{I \cup \{k\}}$, where the sign depends on the number of permutations needed to sort the indices. Therefore, the entry $(M_\omega)_{J,k}$ in row J and column k is given by:

$$(M_\omega)_{J,k} = \begin{cases} \pm p_{J \setminus \{k\}} & \text{if } k \in J, \\ 0 & \text{if } k \notin J. \end{cases}$$

Notice that every entry of M_ω is either 0 or $\pm p_I$. Thus, the entries of M_ω are homogeneous polynomials of degree 1 (linear forms) in the Plücker coordinates.

By Lemmas 1.6 and 1.8, a point $[\omega_0]$ lies in $p(\mathrm{Gr}(d, V))$ iff $\mathrm{rank}(M_{\omega_0}) \leq n - d$. We use the fact that the rank of a matrix equals the size of the largest non-vanishing minor of that matrix, which is equivalent to all $(n-d+1) \times (n-d+1)$ minors of M_{ω_0} vanishing. Each such minor, being the determinant of an $(n-d+1)\times(n-d+1)$ submatrix of M_ω with linear entries, is homogeneous of degree $n - d + 1$ in the Plücker coordinates. Hence $p(\mathrm{Gr}(d, V))$ is the zero locus of a finite collection of homogeneous polynomials in $\mathbb{P}^N$, making it a projective variety. □

Remark 1.10 (Irreducibility via $\mathrm{GL}(V)$). Fix $W_0 = \mathrm{span}\{e_1, \dots, e_d\}$. The group $\mathrm{GL}(V) \cong \mathrm{GL}_n(k)$ acts transitively on $\mathrm{Gr}(d, V)$: any d-dimensional subspace can be mapped to any other by an invertible linear transformation. The natural surjection $\mathrm{GL}_n(k) \to \mathrm{Gr}(d, V)$, $g \mapsto g(W_0)$, descends to a morphism, which is continuous. Since $\mathrm{GL}_n(k)$ is irreducible (it is a principal open subvariety of $\mathbb{A}^{n^2}$), and continuous image of a irreducible set is irreducible, we get that grassmanian is irreducible.

Remark 1.11 (Schemes). The Grassmannian admits a much cleaner definition through which one can realise it as a projective variety. This requires the understanding of schemes. I plan to explore this in the future.

1.5 Example: $\mathrm{Gr}(2, 4)$

For the remainder of this section $\dim V = 4$ with basis $\{e_1, e_2, e_3, e_4\}$. The induced basis of $\bigwedge^2 V$ gives $\mathbb{P}(\bigwedge^2 V) \cong \mathbb{P}^5$ with coordinates $[p_{12} : p_{13} : p_{14} : p_{23} : p_{24} : p_{34}]$. Write $\omega = \sum_{i<j} p_{ij}(e_i \wedge e_j)$. Using the ordered basis $\{e_1 \wedge e_2 \wedge e_3,\ e_1 \wedge e_2 \wedge e_4,\ e_1 \wedge e_3 \wedge e_4,\ e_2 \wedge e_3 \wedge e_4\}$ for $\bigwedge^3 V$, a direct computation of $\varphi_\omega(e_k) = \omega \wedge e_k$ gives the 4×4 matrix (columns indexed by e_1, e_2, e_3, e_4):

$$M_\omega = \begin{pmatrix} p_{23} & -p_{13} & p_{12} & 0 \\ p_{24} & -p_{14} & 0 & p_{12} \\ p_{34} & 0 & -p_{14} & p_{13} \\ 0 & p_{34} & -p_{24} & p_{23} \end{pmatrix}.$$

By Theorem 1.9 with $d = 2$, $n = 4$, $\mathrm{Gr}(2, 4)$ is cut out by all 3×3 minors of M_ω. For example, explicitly computing the minor obtained by removing the fourth row and fourth column yields:

$$\begin{aligned} \begin{vmatrix} p_{23} & -p_{13} & p_{12} \\ p_{24} & -p_{14} & 0 \\ p_{34} & 0 & -p_{14} \end{vmatrix} &= p_{23}(p_{14}^2) - (-p_{13})(-p_{14}p_{24}) + p_{12}(p_{14}p_{34}) \\ &= p_{14}(p_{12}p_{34} - p_{13}p_{24} + p_{14}p_{23}). \end{aligned}$$

Similarly, removing the third row and first column gives:

$$\begin{aligned} \begin{vmatrix} -p_{13} & p_{12} & 0 \\ -p_{14} & 0 & p_{12} \\ p_{34} & -p_{24} & p_{23} \end{vmatrix} &= -p_{13}(p_{12}p_{24}) - p_{12}(-p_{14}p_{23} - p_{12}p_{34}) \\ &= p_{12}(p_{12}p_{34} - p_{13}p_{24} + p_{14}p_{23}). \end{aligned}$$

Notice that each non-trivial 3×3 minor is simply a linear multiple of the same quadratic relation. A straightforward expansion of $\det(M_\omega)$ (which equals, up to sign, the square of the single relation) yields:

Theorem 1.12. $p(\mathrm{Gr}(2,4)) = V(p_{12}p_{34} - p_{13}p_{24} + p_{14}p_{23}) \subset \mathbb{P}^5$.

Proof. **Step 1: image $\subseteq$ variety.** For W with basis $\{w_1 = \sum a_i e_i,\ w_2 = \sum b_i e_i\}$, the Plücker coordinates are $p_{ij} = a_i b_j - a_j b_i$. Substituting:

$$\begin{aligned} p_{12}p_{34} - p_{13}p_{24} + p_{14}p_{23} &= (a_1b_2 - a_2b_1)(a_3b_4 - a_4b_3) \\ &\quad - (a_1b_3 - a_3b_1)(a_2b_4 - a_4b_2) \\ &\quad + (a_1b_4 - a_4b_1)(a_2b_3 - a_3b_2). \end{aligned}$$

Expanding term by term, every monomial $a_i a_j b_k b_l$ appears twice with opposite signs and cancels. Hence the expression is identically zero.

Step 2: variety $\subseteq$ image. Let $[\omega] \in V(p_{12}p_{34} - p_{13}p_{24} + p_{14}p_{23})$ with $\omega \neq 0$. We show ω is totally decomposable. Assume without loss of generality $p_{12} \neq 0$ (all other cases follow by the symmetry of the quadric). Set

$$v_1 = p_{12}\, e_1 + p_{13}\, e_2 + p_{14}\, e_3 + \frac{p_{13}p_{24} - p_{14}p_{23}}{p_{12}}\, e_4, \qquad v_2 = p_{12}\, e_2 + p_{23}\, e_3 + p_{24}\, e_4.$$

Using the quadric relation $p_{12}p_{34} = p_{13}p_{24} - p_{14}p_{23}$, a direct computation gives $v_1 \wedge v_2 = p_{12}\,\omega$. Since $p_{12} \neq 0$ this gives $[\omega] = [v_1 \wedge v_2]$, and $\{v_1, v_2\}$ is a linearly independent set (the e_1-coefficient of v_1 is $p_{12} \neq 0$ while that of v_2 is 0). Thus $[\omega] = p(W)$ for $W = \mathrm{span}\{v_1, v_2\}$. $\square$

Remark 1.13 (Plücker relations and the homogeneous ideal). The polynomials produced in Theorem 1.9 define $\mathrm{Gr}(d, V)$ as a *set*, but they do not in general generate its homogeneous ideal. The actual generators are the *Plücker relations* $\Xi_{\alpha,\beta}(\omega) = \langle {}^t\varphi_\omega(\alpha), {}^t\psi_{\omega^*}(\beta) \rangle$, which are quadratic forms in the Plücker coordinates (see [5] or [6] for the precise construction). The $\mathrm{Gr}(2,4)$ case is deceptively simple; the single quadric $p_{12}p_{34} - p_{13}p_{24} + p_{14}p_{23}$ already generates the ideal. This is special to $\mathrm{Gr}(2,4)$, which has codimension 1 in $\mathbb{P}^5$. For general $\mathrm{Gr}(d,n)$, finding an explicit minimal set of generators for the Plücker ideal, and verifying which minors are redundant is considerably harder; a systematic treatment is carried out in Kolhatkar's thesis [6], where the Plücker ideal, its generators, and the affine charts of the Grassmannian are studied in detail.

2 The Segre Embedding of $\mathbb{P}^m \times \mathbb{P}^n$

2.1 Setup and Motivation

Let U and W be k-vector spaces of dimensions $m+1$ and $n+1$. Identify $\mathbb{P}^m \cong \mathbb{P}(U)$ and $\mathbb{P}^n \cong \mathbb{P}(W)$. Since $\dim(U \otimes W) = (m+1)(n+1)$, we set $N = (m+1)(n+1) - 1$ and write $\mathbb{P}^N = \mathbb{P}(U \otimes W)$. This section establishes that the product of two projective spaces gives us a projective variety, embedded in a much larger space.

2.2 Tensor-Product Definition of Segre Embedding

Definition 2.1 (Segre map).

$$\sigma\colon \mathbb{P}(U)\times\mathbb{P}(W) \longrightarrow \mathbb{P}(U\otimes W), \qquad ([u],[w]) \longmapsto [u\otimes w].$$

Fix bases $\{x_0,\dots,x_m\}$ and $\{y_0,\dots,y_n\}$ for U and W; let Z_{ij} be the dual coordinate on $\mathbb{P}^N$ corresponding to $x_i\otimes y_j$.

Remark 2.2. In the above coordinates, $\sigma([a_0:\dots:a_m],[b_0:\dots:b_n]) = [z_{ij}]$ where $z_{ij}=a_ib_j$. Write $u=\sum_i a_ix_i$ and $w=\sum_j b_jy_j$. Then $u\otimes w=\sum_{i,j} a_ib_j\,(x_i\otimes y_j)$, whose (i,j)-coefficient is precisely a_ib_j. (This is the form of segre map one encounters in the book by Hartshorne [1, Exercise I.2.14].)

2.3 Well-Definedness and Injectivity

Proposition 2.3. *σ is well defined and injective.*

Proof. Well-definedness. Replacing (u,w) by $(\lambda u,\mu w)$ sends $u\otimes w$ to $\lambda\mu(u\otimes w)$; since $\lambda\mu\neq 0$ the projective class is unchanged.

Injectivity. Suppose $[u\otimes w]=\lambda[u'\otimes w']$, i.e. $a_ib_j=\lambda a'_ib'_j$ for all i,j. Fix i_0 with $a_{i_0}\neq 0$; then from $a_{i_0}b_j=\lambda a'_{i_0}b'_j$ for all j we get $(b_j)\propto(b'_j)$, hence $[w]=[w']$. Symmetrically $[u]=[u']$. □

Remark 2.4 (**Categorical product**). The Segre embedding realises $\mathbb{P}^m\times\mathbb{P}^n$ as the *categorical product* in the category of projective varieties: σ satisfies the universal property that any pair of morphisms $f\colon X\to\mathbb{P}^m$ and $g\colon X\to\mathbb{P}^n$ of algebraic varieties factors uniquely through $\mathbb{P}^m\times\mathbb{P}^n$ via (f,g). See [1, Exercise I.3.15] for the general construction of products of varieties and its relation to the Segre embedding.

2.4 Image of the Segre Map and the Defining Ideal

Theorem 2.5. *The image $\Sigma_{m,n}=\operatorname{im}(\sigma)\subset\mathbb{P}^N$ is the projective algebraic set*

$$\Sigma_{m,n} \;=\; V\big(Z_{ij}Z_{kl}-Z_{il}Z_{kj} \;\big|\; 0\le i,k\le m,\ 0\le j,l\le n\big).$$

Proof. Let $Z=(z_{ij})$ be the $(m+1)\times(n+1)$ matrix of homogeneous coordinates. We have the following chain of equivalences:

$$\begin{aligned}
[z_{ij}]\in\operatorname{im}(\sigma) &\iff \exists\, u\in U\setminus\{0\}, w\in W\setminus\{0\} \text{ s.t. } z_{ij}=u_iw_j \text{ for all } i,j\\
&\iff Z=uw^t \quad \text{(where u and w are column vectors)}\\
&\iff \operatorname{rank}(Z)=1 \quad (\text{since } Z\neq 0)\\
&\iff \text{all } 2\times 2 \text{ minors of } Z \text{ vanish}\\
&\iff Z_{ij}Z_{kl}-Z_{il}Z_{kj}=0 \quad \text{for all } i,j,k,l.
\end{aligned}$$

Since each converse of each implication holds true, the polynomials strictly generated by the vanishing $2{\times}2$ minors are exactly what we need to completely define $\operatorname{im}(\sigma)$. □

Example 2.6 (The specific case $\mathbb{P}^1 \times \mathbb{P}^2$)**.** Let $m = 1$ and $n = 2$. The coordinate matrix is given by:

$$Z = \begin{pmatrix} z_{00} & z_{01} & z_{02} \\ z_{10} & z_{11} & z_{12} \end{pmatrix}.$$

Following the exact logic established in the general case, $\operatorname{rank}(Z) = 1$ if and only if all of its 2×2 minors vanish. There are exactly three such minors:

$$F_1 = z_{00}z_{11} - z_{01}z_{10}, \quad F_2 = z_{00}z_{12} - z_{02}z_{10}, \quad F_3 = z_{01}z_{12} - z_{02}z_{11}.$$

Thus, $\Sigma_{1,2} = \operatorname{im}(\sigma) = V(F_1, F_2, F_3) \subset \mathbb{P}^5$.

2.5 Irreducibility and the Projective Nullstellensatz

Theorem 2.7. *The algebraic set $\Sigma_{m,n}$ is irreducible, and its homogeneous ideal is given exactly by $I(\Sigma_{m,n}) = (Z_{ij}Z_{kl} - Z_{il}Z_{kj})$.*

Proof. Consider the polynomial ring homomorphism

$$\phi\colon k[Z_{ij}] \longrightarrow k[x_0, \dots, x_m, y_0, \dots, y_n], \qquad Z_{ij} \longmapsto x_i y_j.$$

The image of ϕ is the subring $R = k[x_i y_j \mid 0 \le i \le m,\, 0 \le j \le n]$. By the First Isomorphism Theorem, we know that $k[Z_{ij}]/\ker\phi \cong R$. Since R embeds naturally as a subring into the integral domain $k[x_0, \dots, x_m, y_0, \dots, y_n]$, R itself must be an integral domain. Therefore, $\ker\phi$ is a prime ideal, and its zero locus $V(\ker\phi)$ defines an irreducible projective variety.

We now identify $V(\ker\phi)$. The quadrics $Z_{ij}Z_{kl} - Z_{il}Z_{kj}$ evaluate to zero under ϕ (since $(x_i y_j)(x_k y_l) - (x_i y_l)(x_k y_j) = 0$), so they lie in $\ker\phi$. Consequently, $V(\ker\phi) \subseteq V(Z_{ij}Z_{kl} - Z_{il}Z_{kj}) = \Sigma_{m,n}$. Furthermore, any point $[z_{ij}] \in \Sigma_{m,n}$ clearly satisfies the image definition $z_{ij} = x_i y_j$, hence residing in $V(\ker\phi)$. Thus, $V(\ker\phi) = \Sigma_{m,n}$.

Applying the Projective Nullstellensatz, the homogeneous ideal of the variety is $I(\Sigma_{m,n}) = \operatorname{rad}(\ker\phi)$. Because $\ker\phi$ is a prime ideal, it is already equal to its own radical. This immediately implies that $I(\Sigma_{m,n}) = \ker\phi$. $\square$

Example 2.8 (Irreducibility for $\mathbb{P}^1 \times \mathbb{P}^2$)**.** Specialising to $m = 1, n = 2$, the homomorphism $\phi\colon k[Z_{00}, \dots, Z_{12}] \to k[x_0, x_1, y_0, y_1, y_2]$ has a kernel containing the ideal generated by the three specific minors (F_1, F_2, F_3). Because $k[Z_{ij}]/\ker\phi \cong R$ is an integral domain, $\ker\phi$ is prime. As reasoned above, $V(\ker\phi) = \Sigma_{1,2}$. This directly asserts that $\Sigma_{1,2}$ is a projective variety (embedded in $\mathbb{P}^5$), and its homogeneous ideal is $I(\Sigma_{1,2}) = \ker\phi = (F_1, F_2, F_3)$.

Bibliography

[1] R. Hartshorne, *Algebraic Geometry*, Graduate Texts in Mathematics 52, Springer-Verlag, 1977.

[2] J. Harris, *Algebraic Geometry: A First Course*, Graduate Texts in Mathematics 133, Springer-Verlag, 1992.

[3] I. R. Shafarevich, *Basic Algebraic Geometry 1: Varieties in Projective Space*, 3rd ed., Springer, 2013.

[4] P. Griffiths and J. Harris, *Principles of Algebraic Geometry*, Wiley-Interscience, 1978.

[5] D. A. Hudec, *The Grassmannian as a Projective Variety*, University of Chicago REU, 2007.

[6] R. Kolhatkar, *Grassmann Varieties*, M.Sc. thesis, McGill University, 2004.